Interactions
User Interfaces

Capstone Project

Maura Cerioli

User Interaction ⇔ User Interface

For most users Interface ≅ System

- ✓features
- ✓ease of use
- ×performances
  - besides interface responsiveness
- ×security/safety
  - besides the obvious

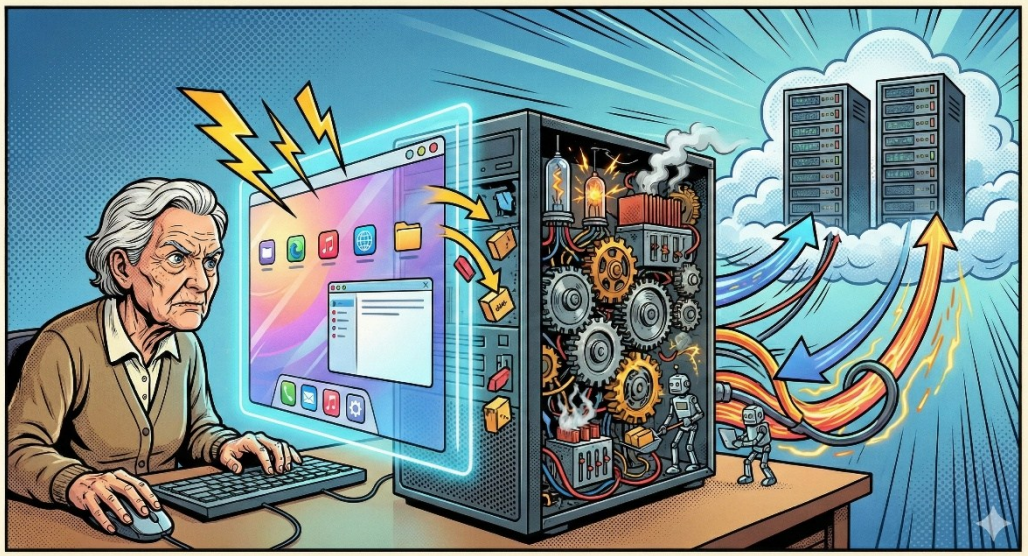
Improving interface

dramatic improvement of perceived quality

User



Interface determines user interactions





Limited by available components

Flexible and easier to creatively design



Dibris

Usability ⇔ Accessibility ⇔ User Experience



Usability

ease of use + learnability for specified users



Accessibility

availability to as many people as possible

User Experience
≅
usability + feelings



User Experience

A person's perceptions and responses that result from the use or anticipated use of a product, system or service

(also short for [good] user experience design)

Dibris

Usability

One or more metric for each facet
Categories for testing/measuring usability
Test = user observation

Facets

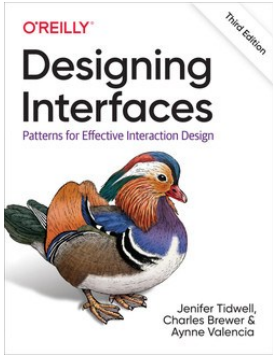
- Learnability
- Efficiency
- Memorability
- Errors
- Satisfaction

Get the *right* set of user
Sets them some tasks
Measure results

- # successes
- # errors
- # time to completion
- ….

Jakob Nielsen & Ben Shneiderman

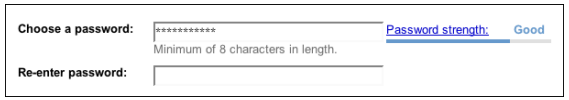
Interaction Patterns



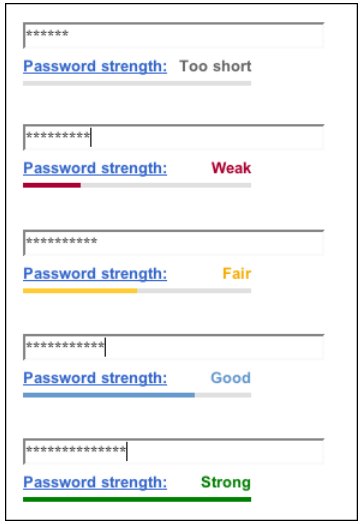
Many books/sites with catalogues

- sometimes listing design concepts that are not exactly design patterns
- e.g.,
  - https://ui-patterns.com/patterns
  - https://uxplaybook.org/articles/essential-interaction-design-patterns-and-techniques
  - https://www.smashingmagazine.com/2009/06/10-ui-design-patterns-you-should-be-paying-attention-to/
- find out more on your own
  - it might be your secret weapon for the project
- many examples seen web surfing
  - still useful to bring them to conscious level

Password Strength Meter



- What: give the user immediate feedback on the validity and strength of a new password while it is being typed
- Use when: the UI asks the user to choose a new password
- Why: strong passwords protect both users and site
  - weak passwords ought to be disallowed
  - immediate feedback
    - instantly, not after form submission
- How: while user typing/as focus leaves the text field, show an estimate of the password strength beside the text field.
  - at minimum, text /graphic label indicating a weak, medium, or strong password + wording to describe weakness, if any
  - if possible, additional specific advice on how to produce a strong password
  - never actually show a password/make suggestions of alternative passwords
- Examples

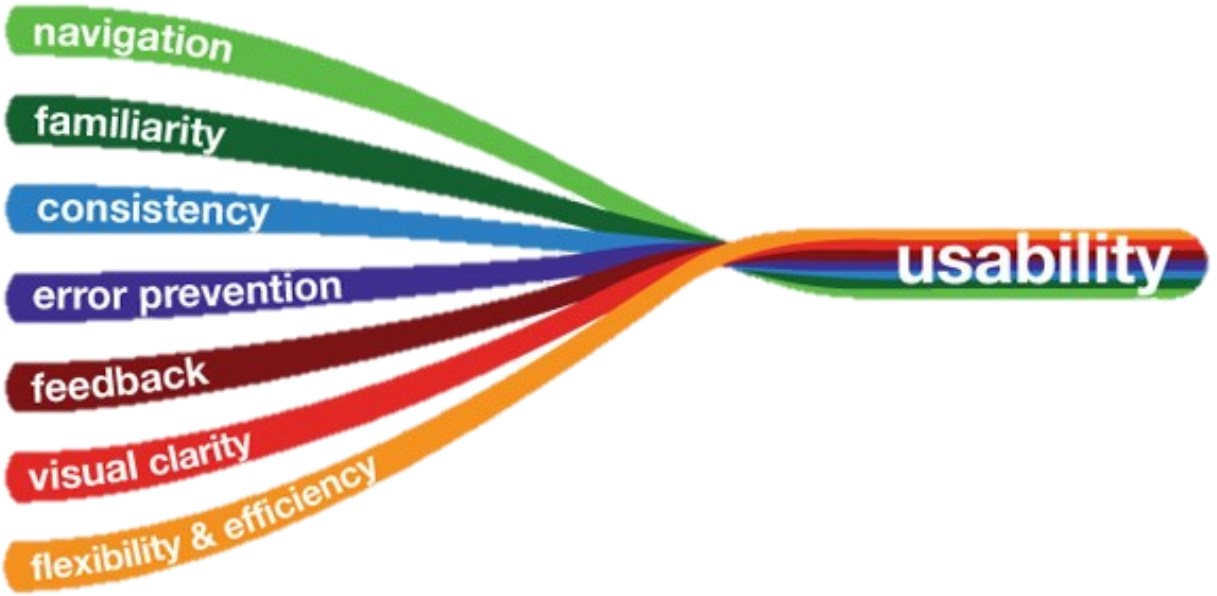




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Usable…but HOW?

- Several lists of
  - tips
  - areas of concern
- Based on results from cognitive science
  - human physiology
  - human standard behaviour



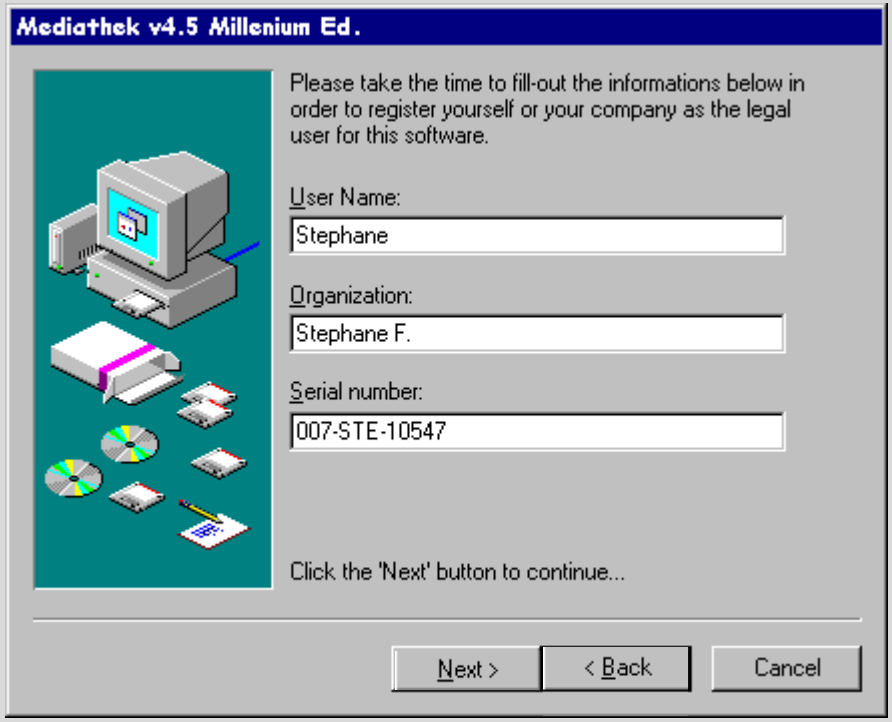
Facts about Human Perception & Cognition

1. We perceive what we expect
2. Our vision is optimized to see structure
3. Our color vision is limited
4. Our peripheral vision is poor
5. Our attention is limited
6. Our memory is imperfect
7. Recognition is easy; recall is hard
8. We think mostly about our tasks, not our tools
9. We seek and use structure
10. Inductive reasoning is easy; deduction is hard
11. Human thought-cycle: goal, execute, evaluate
12. We have real-time requirements

Jeff Johnson

Facts about Human Perception & Cognition

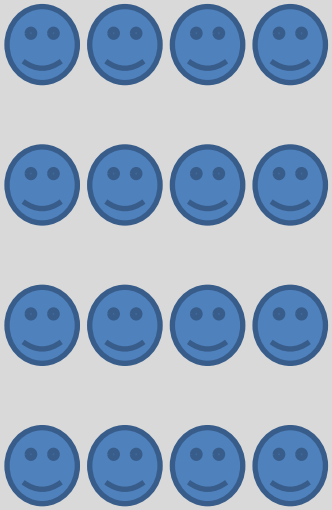
1. We perceive what we expect

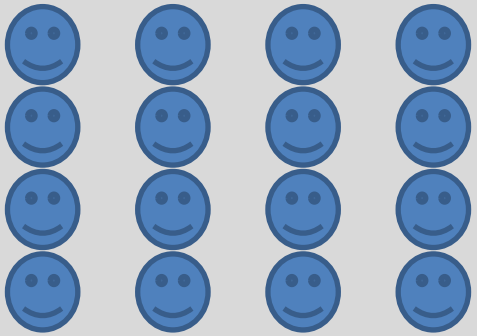


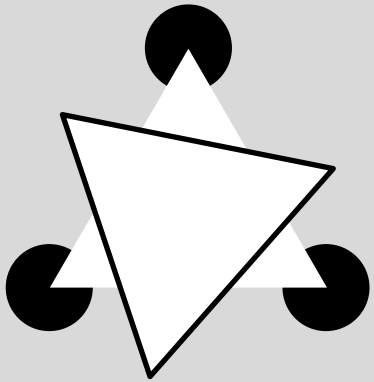
Dibris

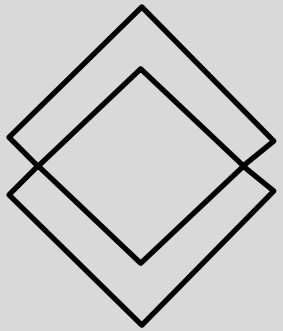
Facts about Human Perception & Cognition

2. Our vision is optimized to see structure





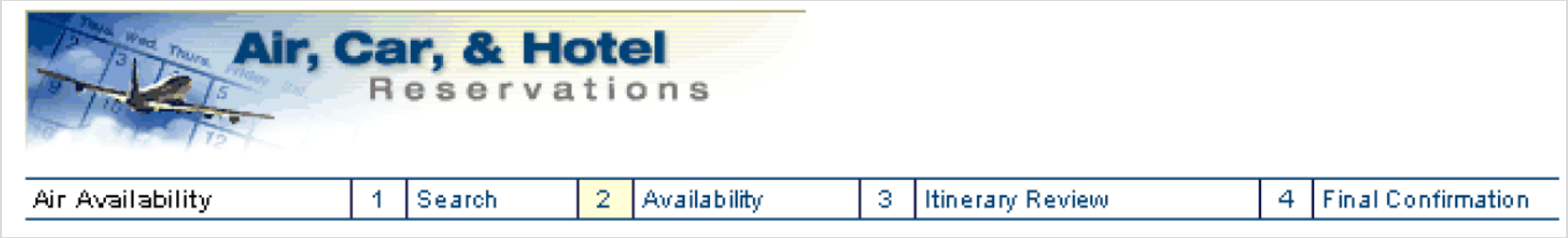




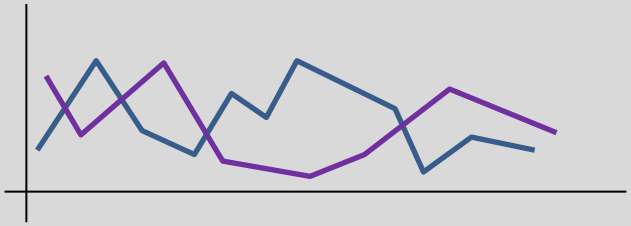
Facts about Human Perception & Cognition

3. Our color vision is limited

```
30
31    test 'should not multiple create posts' do
32      assert_difference('Post.count') do
33        post :create, post: @updated_post
34      end
35      assert_no_difference('Post.count') do
36        post :create, post: @updated_post
37      end
38    end
```



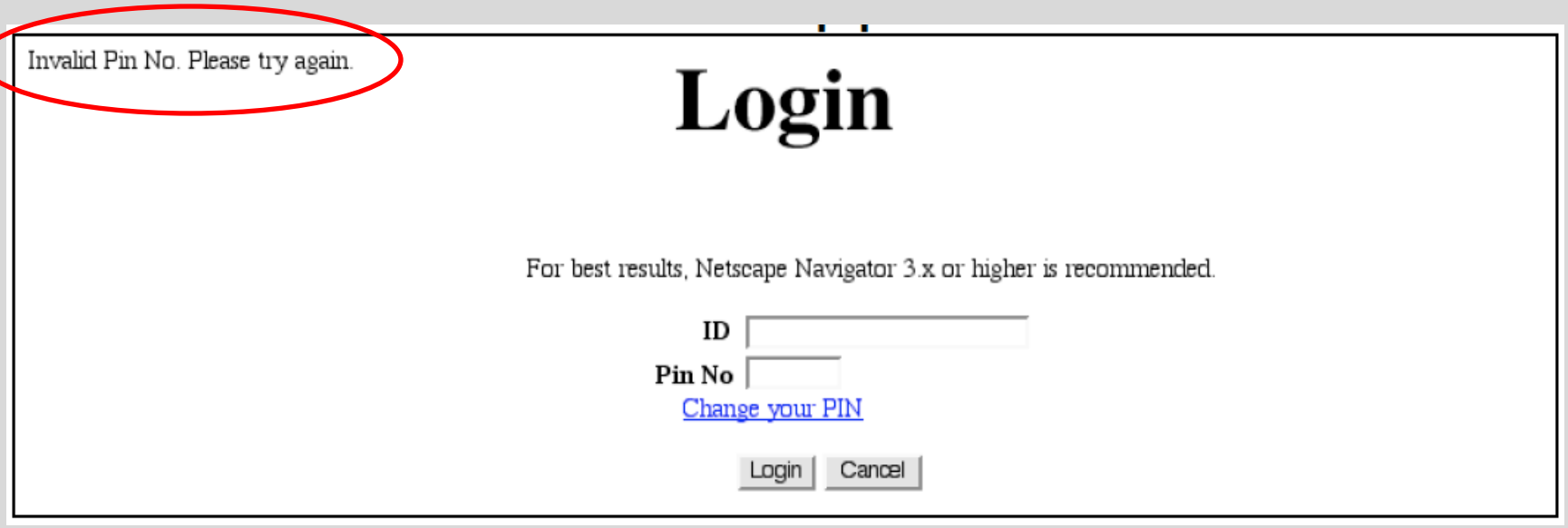
colour blindness: ~8% male, ~0.5% female



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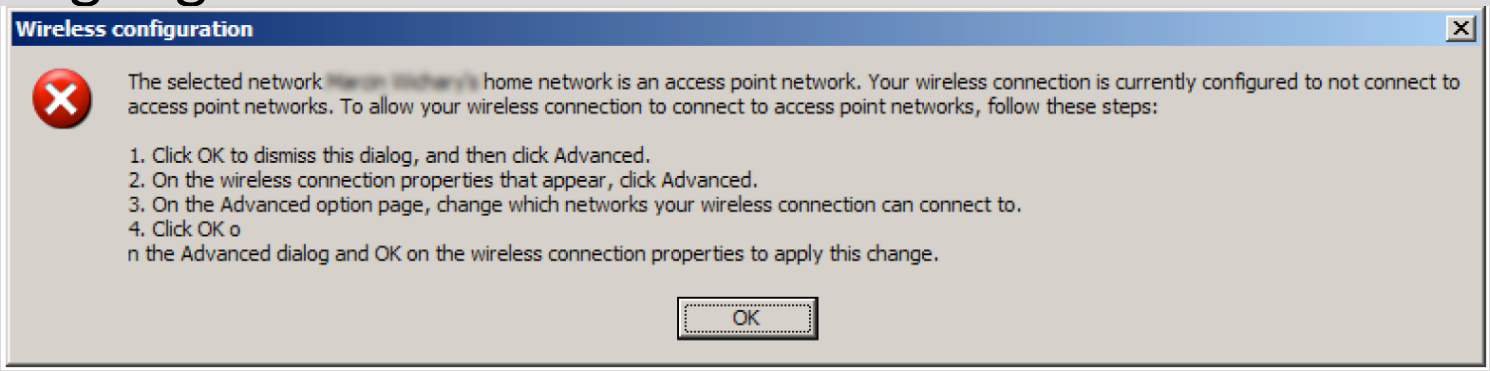
Facts about Human Perception & Cognition

4. Our peripheral vision is poor



Facts about Human Perception & Cognition

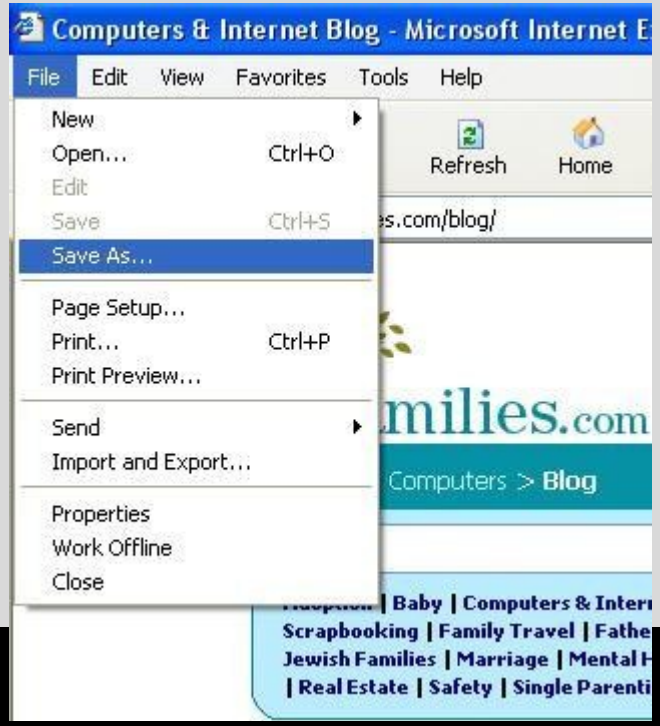
5. Our attention is limited
6. Our memory is imperfect
   Short-term memory (STM)
   - Represents conscious mind
     - Attention: what we're attending to NOW
   - Holds 3-7 unrelated items
   - Goals, numbers, words, images, other info
   - New items can "bump" old ones out
   - Easy to forget goals or info



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Facts about Human Perception & Cognition

7. Recognition is easy; recall is hard

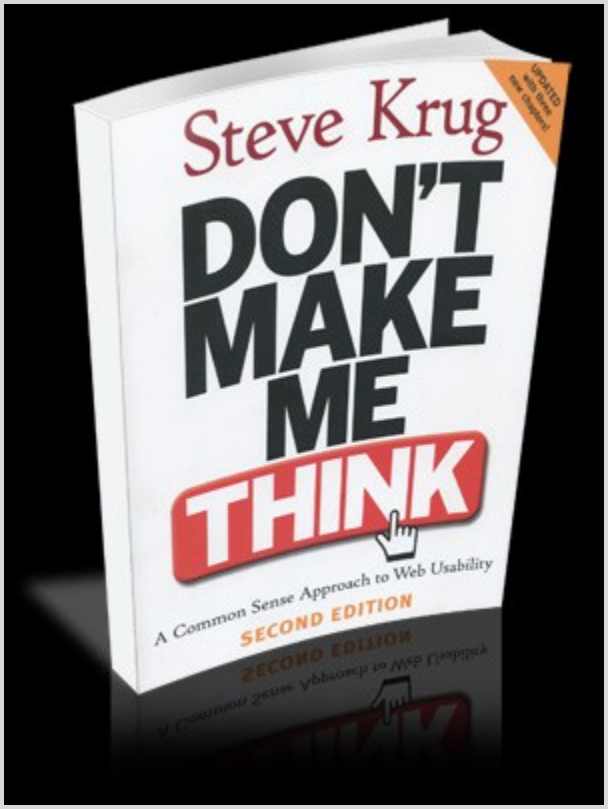




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Facts about Human Perception & Cognition

8. We think mostly about our tasks, not our tools



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Facts about Human Perception & Cognition

9. We seek and use structure

Create a Clear Visual Hierarchy
Organize and prioritize the contents of a page by using size, prominence, and content relationships. Let's look at these relationships more closely.
The more important a headline is, the larger its font size should be.
Big bold headlines help to grab the user's attention as they scan the Web page. The more important the headline or content, the higher up the page it should be placed.
The most important or popular content should always be positioned prominently near the top of the page, so users can view it without having to scroll too far. Group similar content types by displaying the content in a similar visual style, or in a clearly defined area.

Dibris

Create a Clear Visual Hierarchy

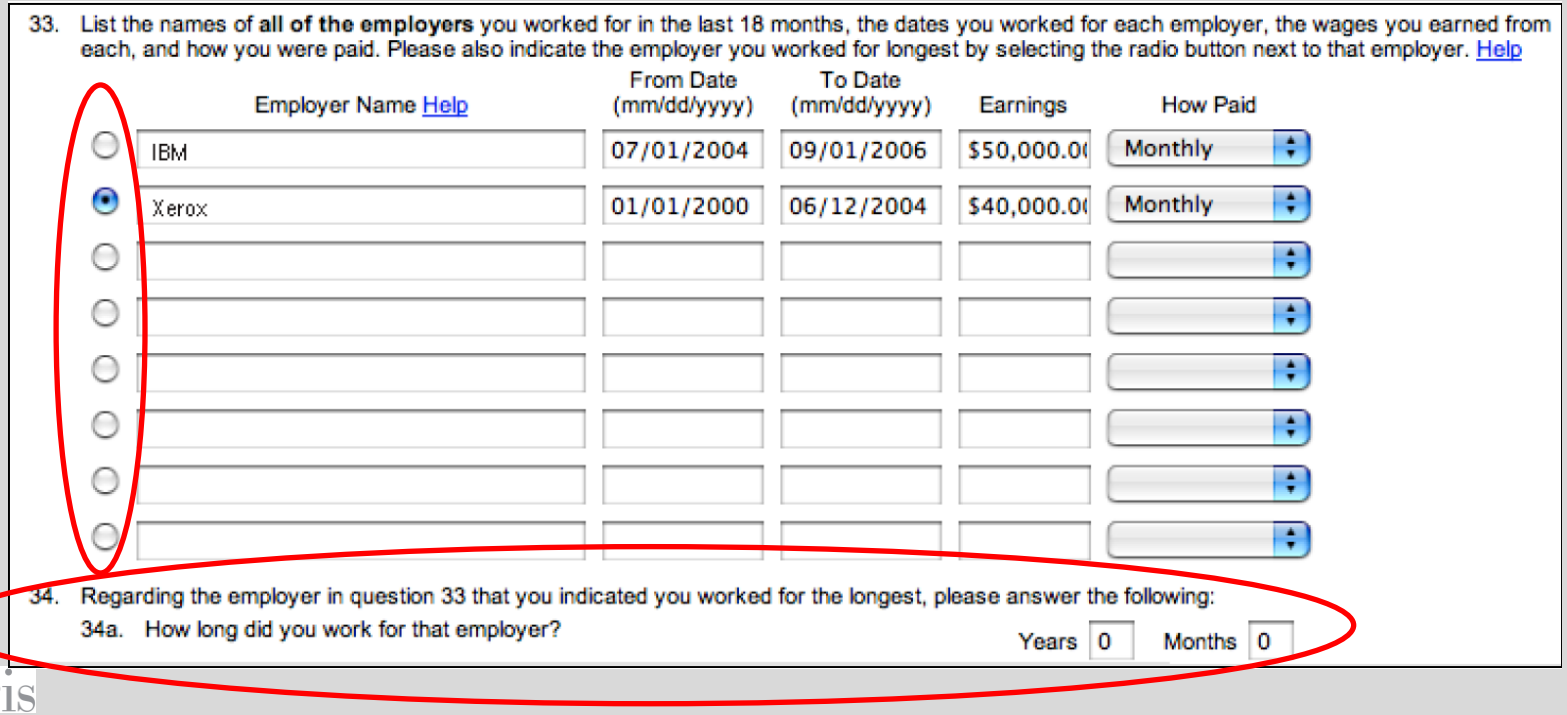
Organize and prioritize the contents of a page by using size, prominence, and content relationships.

Let's look at these relationships more closely:

- **Size**. The more important a headline is, the larger its font size should be. Big bold headlines help to grab the user's attention as they scan the Web page.
- **Prominence**. The more important the headline or content, the higher up the page it should be placed. The most important or popular content should always be positioned prominently near the top of the page, so users can view it without having to scroll too far.
- **Content Relationships.** Group similar content types by displaying the content in a similar visual style, or in a clearly defined area.

Facts about Human Perception & Cognition

10. Inductive reasoning is easy; deduction is hard
    = learning from experience is natural, making calculations is not
    ⇒ don't make users deduce/compute unless strictly necessary



33. List the names of **all of the employers** you worked for in the last 18 months, the dates you worked for each employer, the wages you earned from each, and how you were paid. Please also indicate the employer you worked for longest by selecting the radio button next to that employer. Help

Employer Name Help	From Date (mm/dd/yyyy)	To Date (mm/dd/yyyy)	Earnings	How Paid
IBM	07/01/2004	09/01/2006	$50,000.0(	Monthly
Xerox	01/01/2000	06/12/2004	$40,000.0(	Monthly

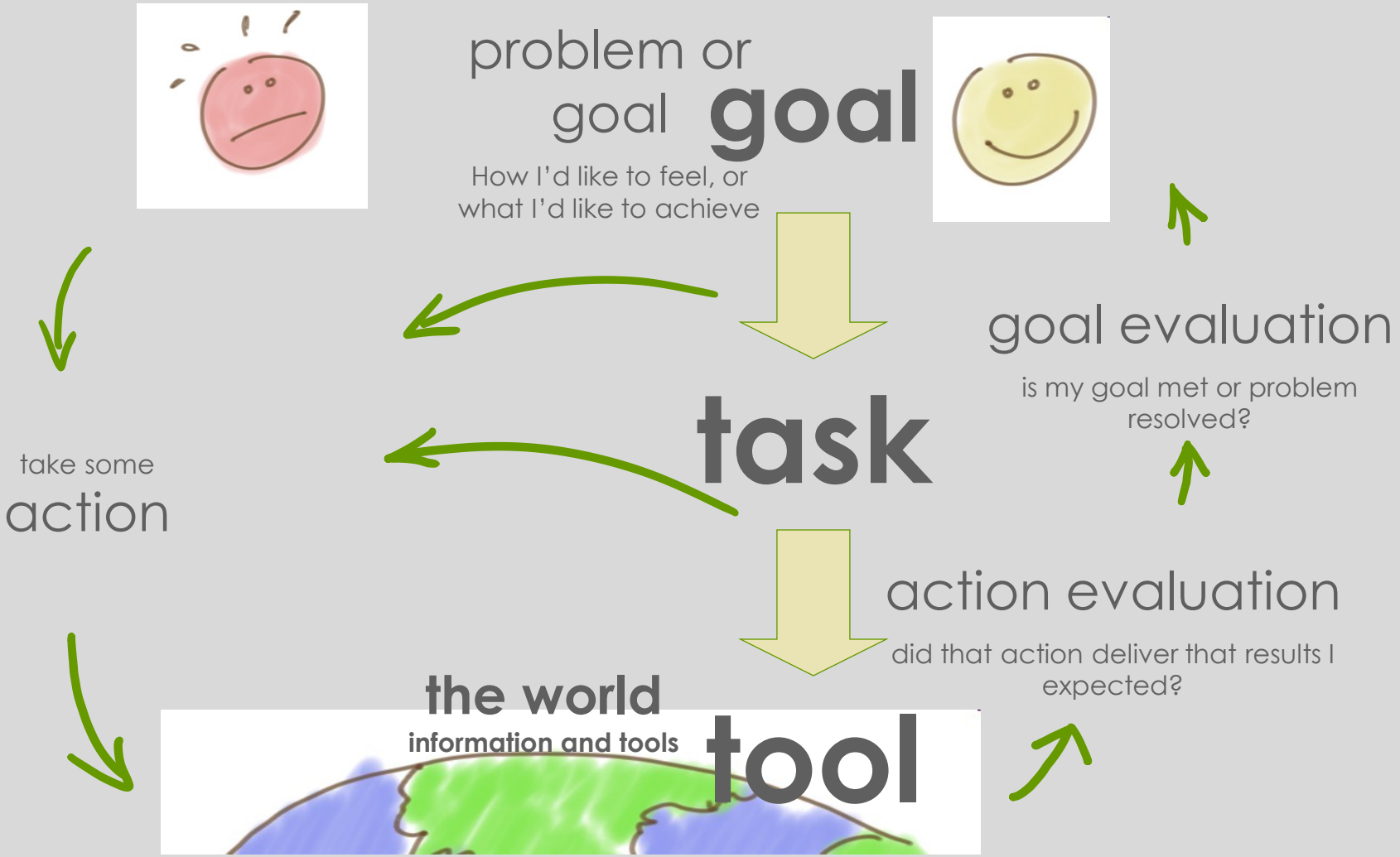
34. Regarding the employer in question 33 that you indicated you worked for the longest, please answer the following:

34a. How long did you work for that employer? Years 0 Months 0

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Facts about Human Perception & Cognition

11. Human thought-cycle: goal, execute, evaluate (Norman's model)



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Facts about Human Perception & Cognition

12. We have real-time requirements
    - 0.1 Second - Governs human
        - Perceptual-motor time-loop
        - Perception of cause-effect
        - Perceptual fusion
    - 1 Second - Governs human
        - Turn-taking in conversation
        - Minimum reaction time for unexpected events
    - 10 Seconds - Governs human
        - Unbroken attention to a task
        - “Unit task”







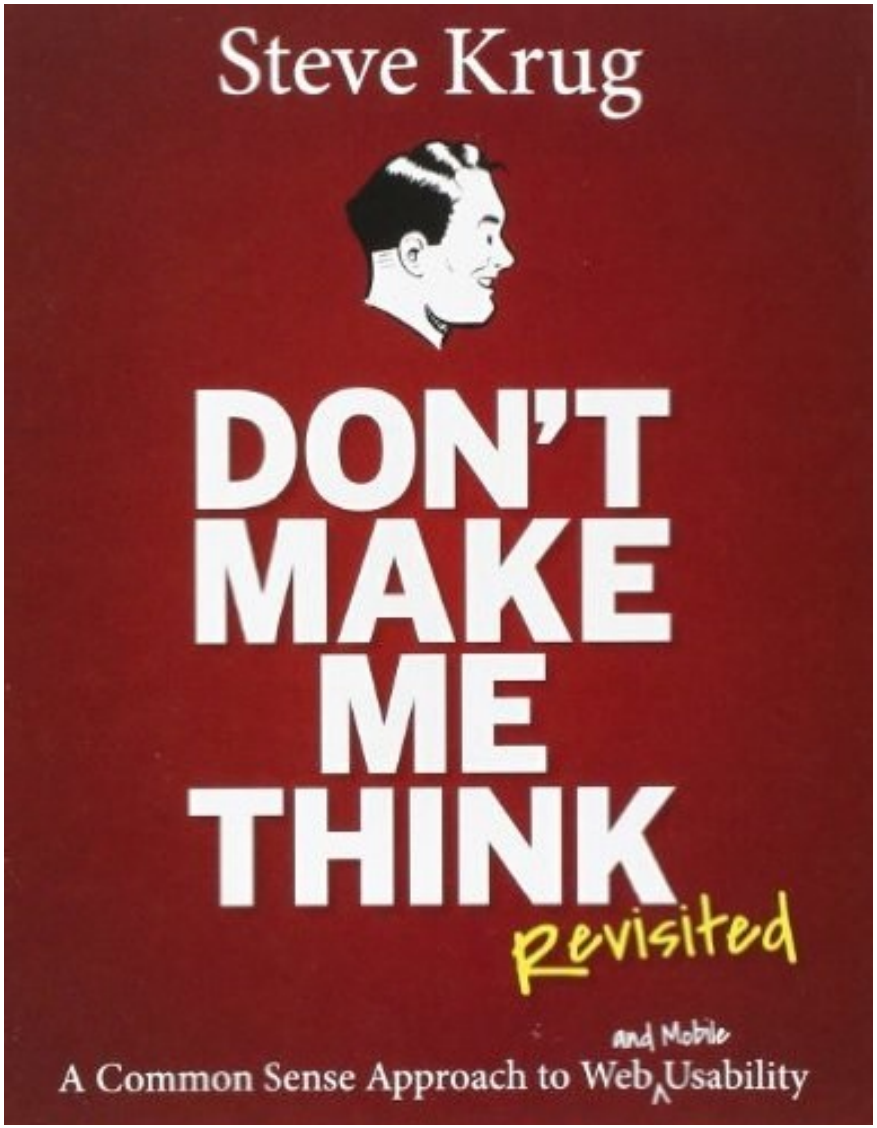




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User interface design principles

Principle	Description
User familiarity	The interface should use terms and concepts which are drawn from the experience of the people who will make most use of the system.
Consistency	The interface should be consistent in that, wherever possible, comparable operations should be activated in the same way.
Minimal surprise	Users should never be surprised by the behaviour of a system.
Recoverability	The interface should include mechanisms to allow users to recover from errors.
User guidance	The interface should provide meaningful feedback when errors occur and provide context-sensitive user help facilities.

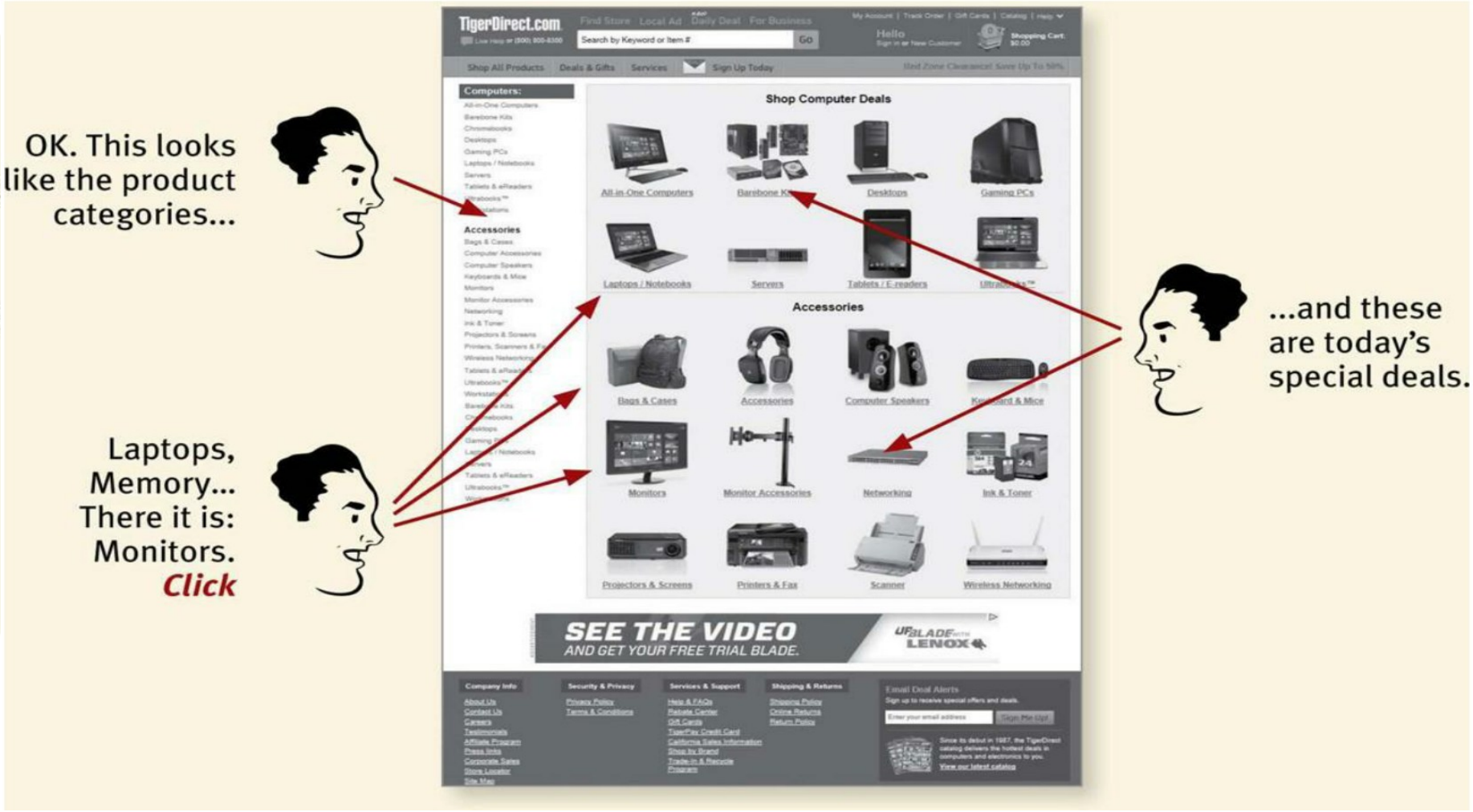


Steve Krug - New Riders Pub
2nd revised edition
26th of August 2005

Steve Krug - New Riders Pub
3rd revised edition
24th of December 2013

Addresses also Mobile Apps

Think? in Which Sense?

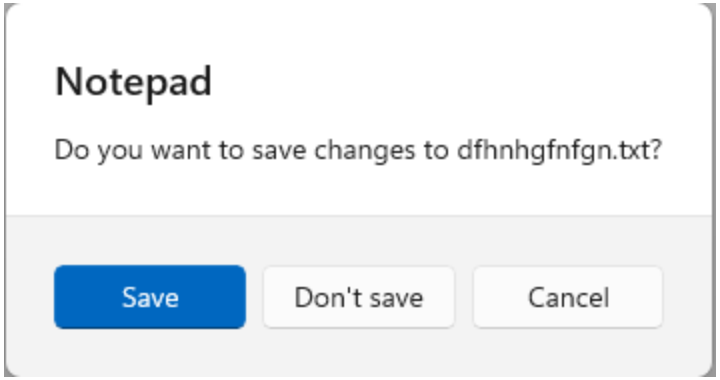


UX designer goal: getting rid of those ???

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Other examples







Making every detail self-evident is like having good lighting in a store it just makes everything seem better.

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A few (non sw) Design Disasters





userinyerface.com





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Designers' Expectations vs. Reality

WHAT WE DESIGN FOR...

Read

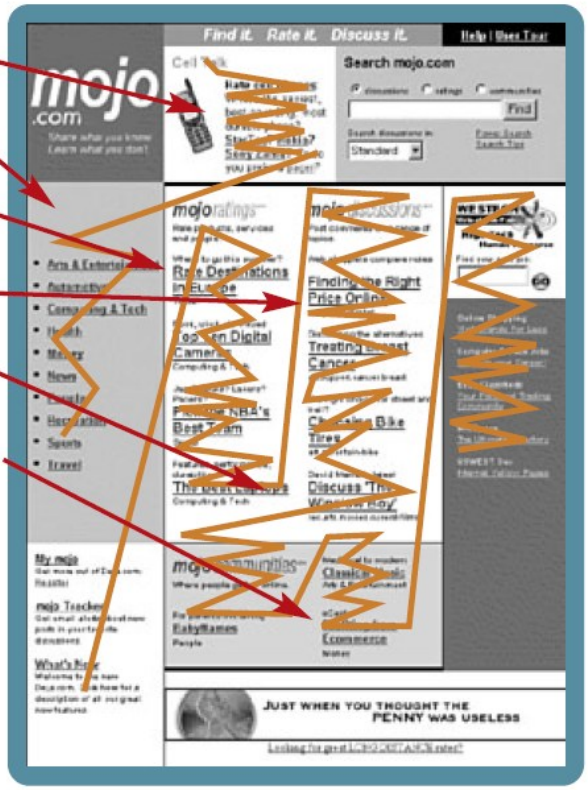
Read

Read

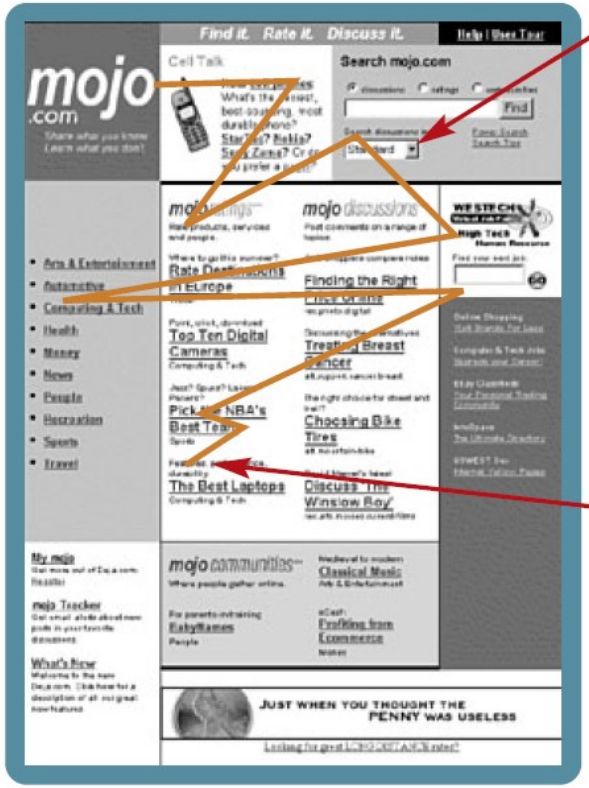
Read

[Pause for reflection]

Finally, click on a carefully chosen link



THE REALITY...



Look around feverishly for anything that

a) is interesting, or vaguely resembles what you're looking for, and

b) is clickable.

As soon as you find a halfway-decent match, click.

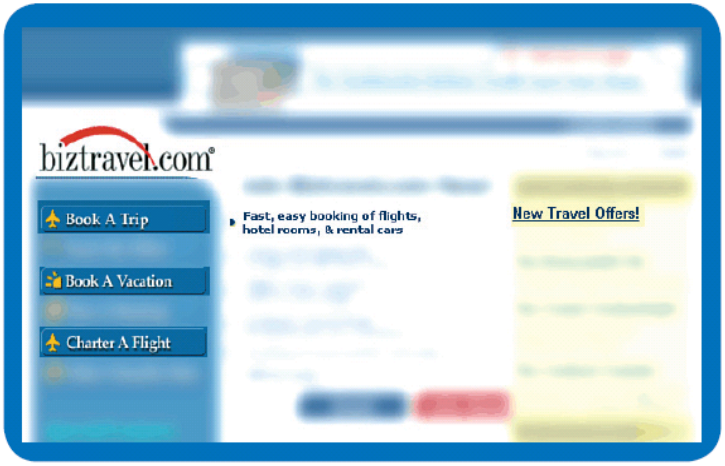
If it doesn't pan out, click the Back button and try again.

Goal Oriented Scanning

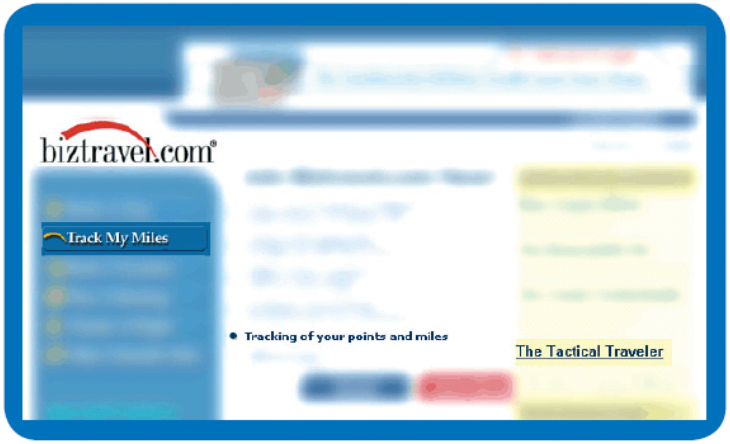




I want to buy a ticket



FACT OF LIFE #1
We don't read pages.
We scan them



How do I check my frequent flyer's miles?



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Scanning…Is It Real? (heat maps)

- It is
  Tracking eye's focus
  - red areas highest number of hits, then yellow, then blue
- F shaped = reading stops in the middle of sentences



FACT OF LIFE #2
We don't make optimal choices.
We satisfice.

- Satisficing = avg(satisfying, sufficing)
- Why not looking for the best choice?
  - Hurry: "Optimizing is hard, and it takes a long time. Satisficing is more efficient."
  - Not much of a penalty for guessing wrong: a click or two of the Back button
  - Weighing options may not improve our chances on poorly designed interfaces
  - More fun: easier + chance of running into something surprising and good

FACT OF LIFE #3
We don't figure out how things work
We muddle through

- What's a browser?
  - a search tool
  - google



- People don't care to understand as far as it works
- If we find something that works, we stick to it

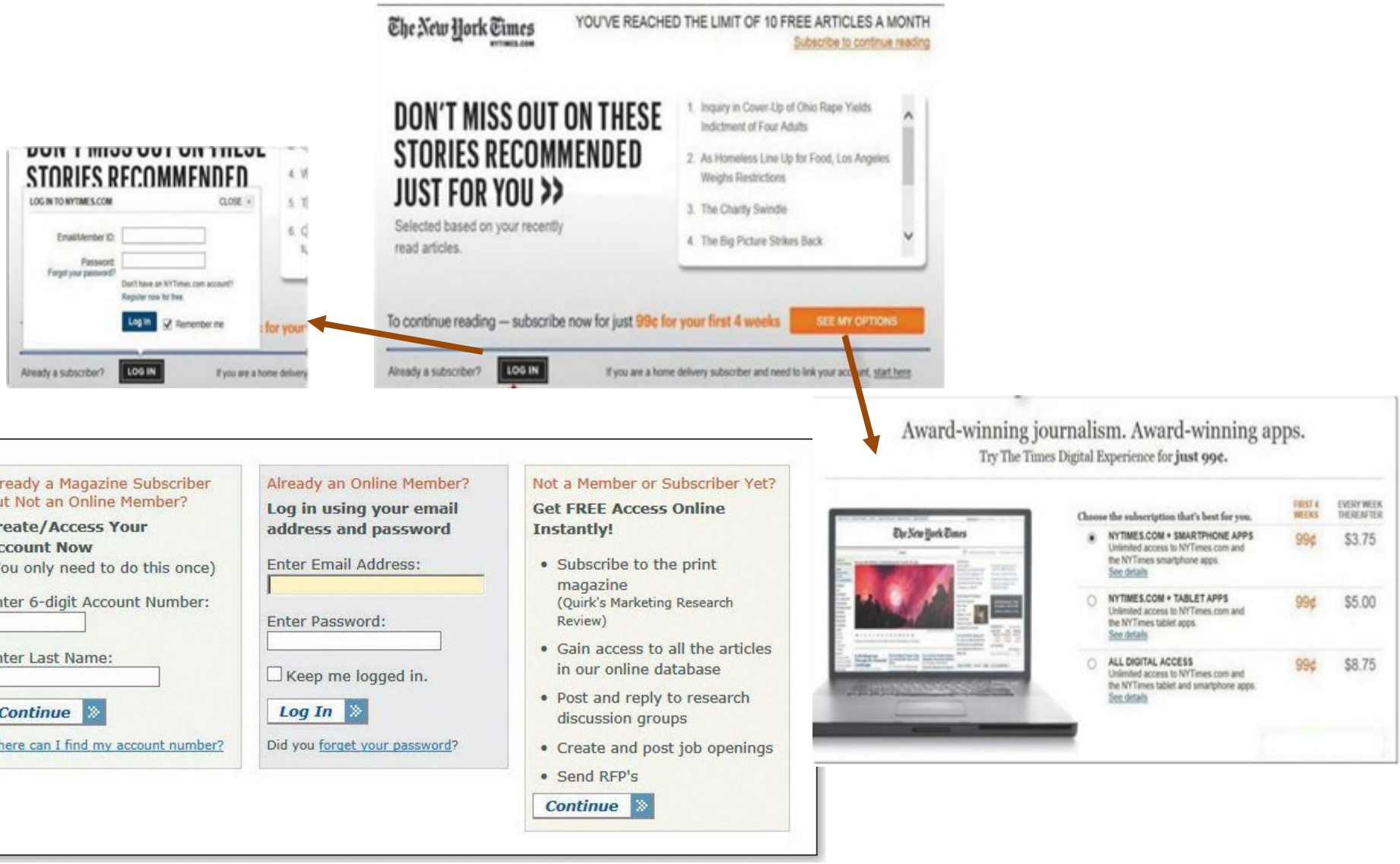
Law#1: Don’t Make Me Think
Designing for Scanning - not Reading

- Take advantage of conventions
- Create effective visual hierarchies
- Break pages up into clearly defined areas
- Make it obvious what’s clickable
- Eliminate distractions
- Format content to support scanning

Law #2: It doesn't matter how many times I have to click, as long as each click is a mindless, unambiguous choice

- Costs of a click
  - load page time
  - effort for understanding/making a decision
  - decision time
- In most cases 3 brainless clicks = 1 motivated click

Making Clicks Painless: an Example

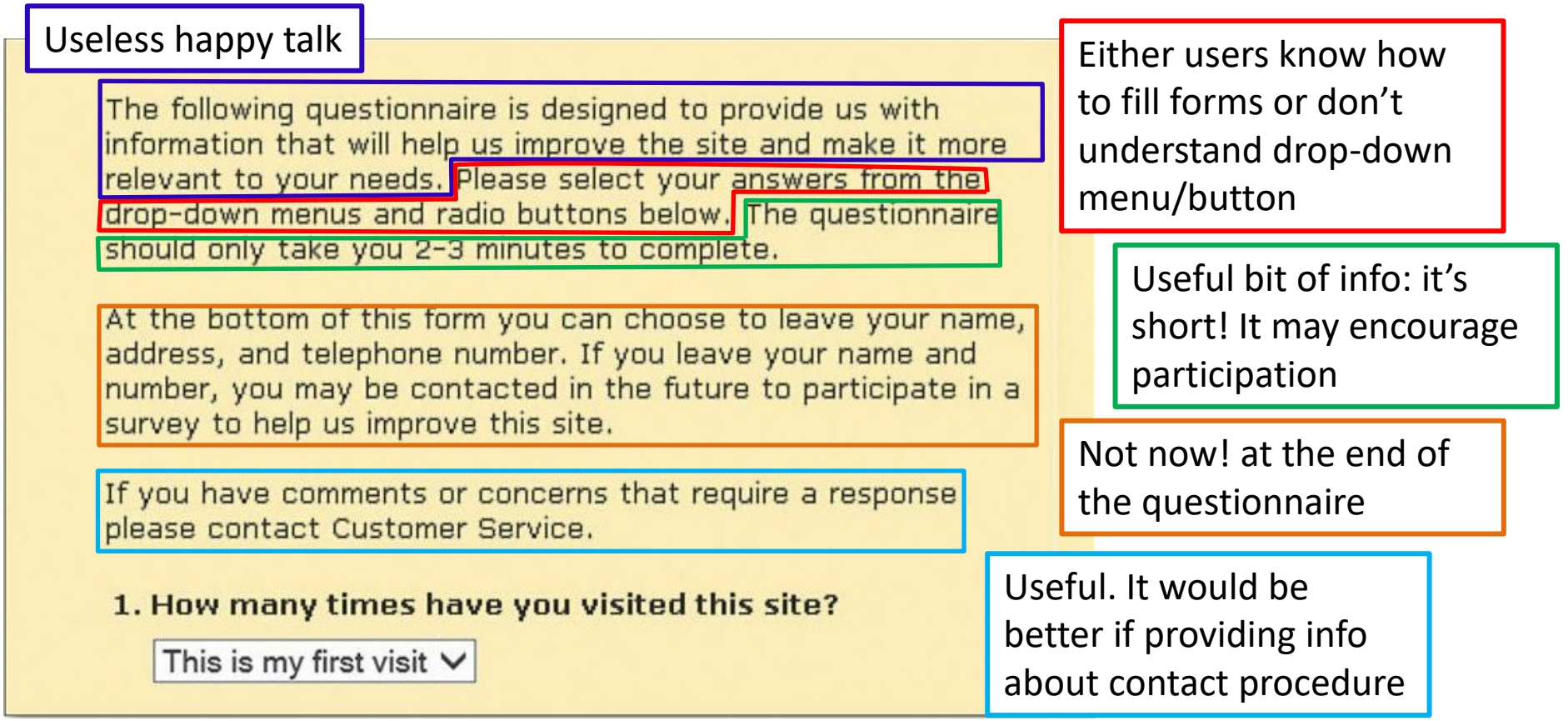


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Law #3: Get rid of half the words on each page then get rid of half of what's left

- Pros
  - Lower noise level
  - Contents more prominent
  - Pages shorter ⇒ users see more without scrolling
- How
  - Happy talk must die
    users are not here for tea
  - Instructions must die
    everything should be self-explanatory

Merciless Word Pruning: an Example



Please help us improve the site by taking 2-3 minutes to complete this survey.

NOTE: If you have comments or concerns that require a response, don't use this form. Instead, please contact Customer Service.

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Summary

DON'T MAKE ME THINK

A COMMON SENSE APPROACH TO WEB USABILITY

by STEVE KRUG



"MANY PEOPLE WHO ENCOUNTER PROBLEMS WITH A SITE TEND TO BLAME THEMSELVES AND NOT THE SITE."

THE USABILITY LAWS

1. DON'T MAKE ME THINK!
2. IT DOESN'T MATTER HOW MANY TIMES I HAVE TO CLICK, AS LONG AS EACH CLICK IS A MINDLESS, UNAMBIGUOS CHOICE.
3. GET RID OF HALF THE WORDS ON EACH PAGE, THEN GET RID OF HALF OF WHATS LEFT.

3 FACTS OF WEB LIFE

1. WE DON'T READ PAGES. WE SCAN THEM.
2. WE DON'T MAKE OPTIMAL CHOICES. WE SATISFICE.
3. WE DON'T FIGURE OUT HOW THINGS WORK. WE MUDDLE THROUGH.

How to know if we did things right?

- No way to define automated tests from specifications
  - how to automate sentiments/feelings?
- We need feedback from users $\Rightarrow$ User testing
- Developers are users, wouldn't their opinion suffice?
  - no, if project for specific type of users
  - not the standard users even for people at large
- OK, but developers know how "generic" people are
  - aka "my mom wouldn't like/understand *syndrome*"

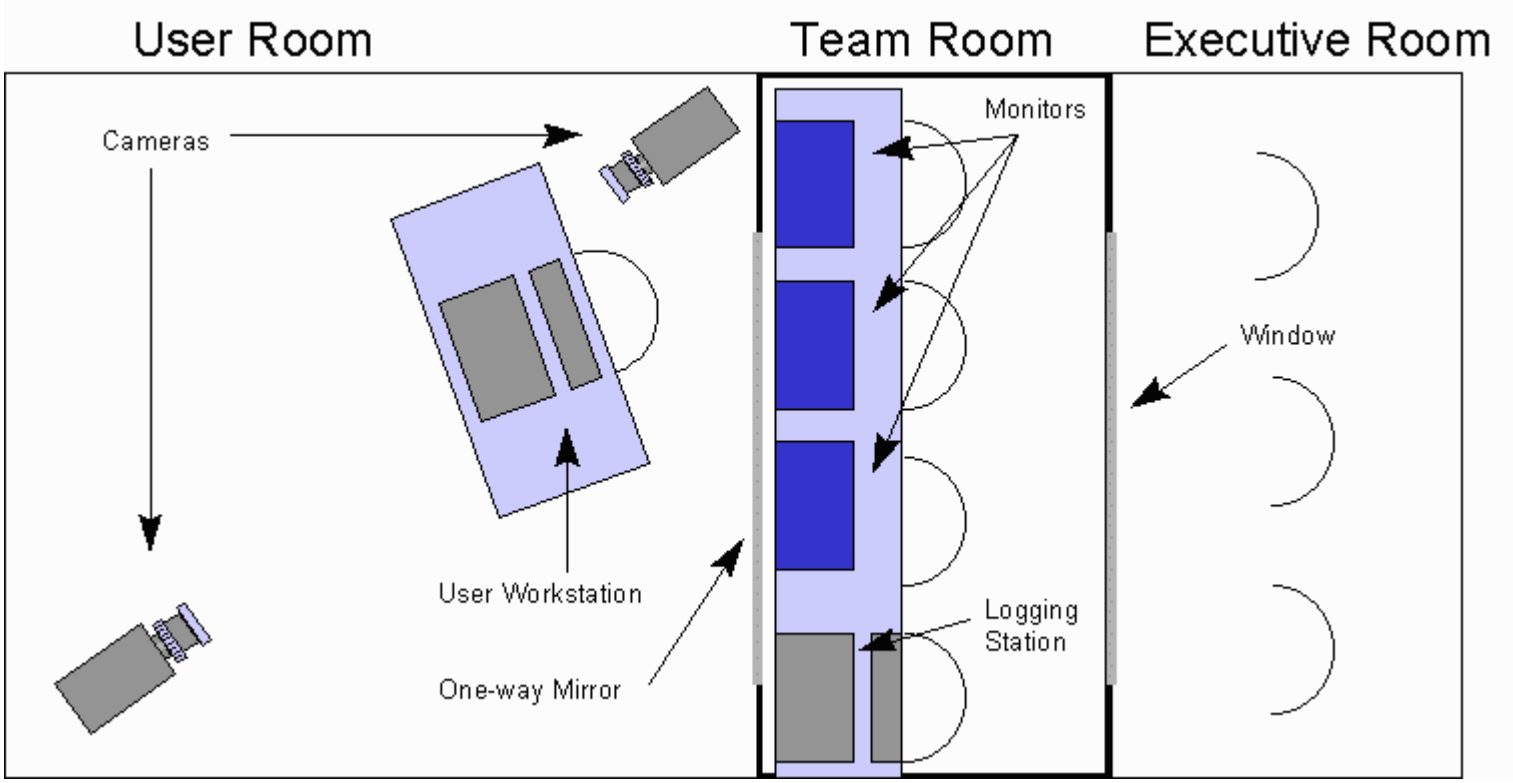
In doubt? Ask the Users!

- Usability Testing
  = watching people trying to use the system while thinking out loud
- Arguments are impartially ruled
  - watching users
  - no personal biases
- Shared experience
- Aha moment: *our users are not like us*
  - see the system through their eyes





Usability Testing



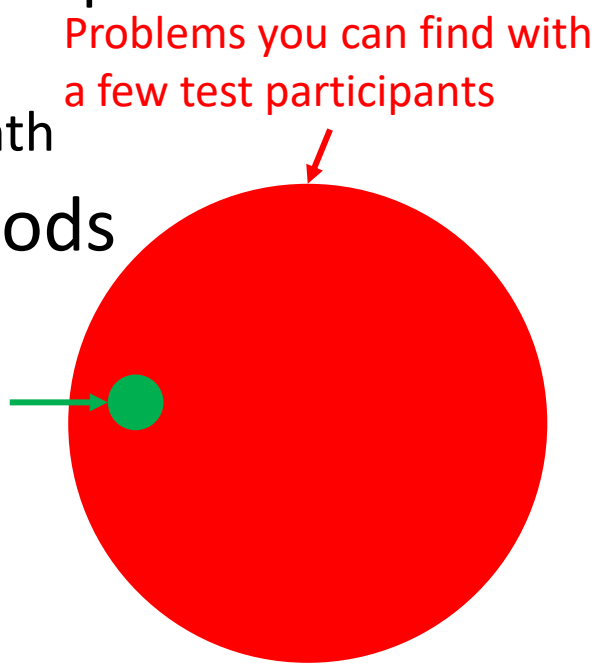
- Lab + experienced professional
  - Same script for several users
    - 8 minimum
  - Record session for further analysis
  - weeks of work
- Analytical report of findings



$5k-$10K

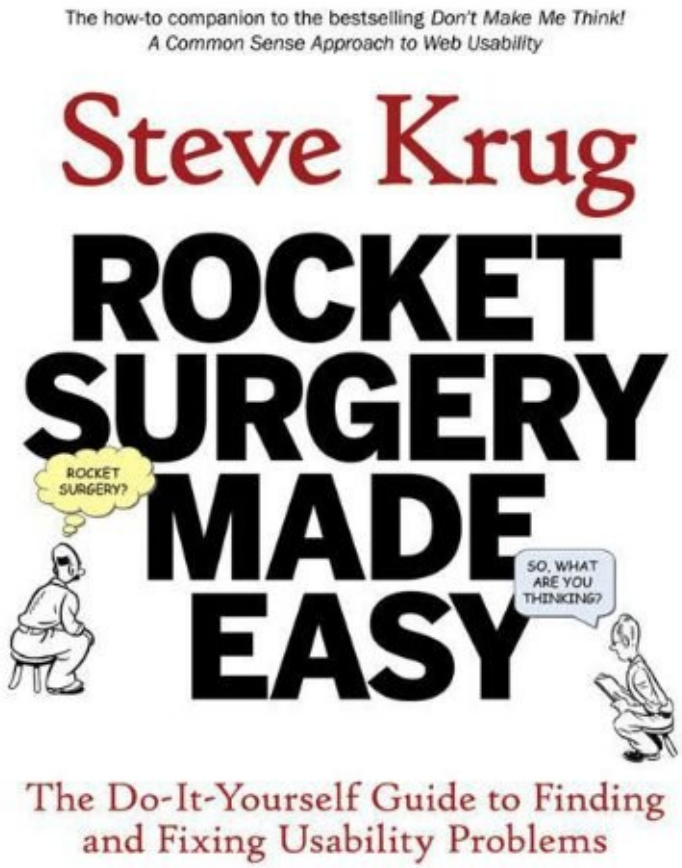
Usability Testing Works

- Uncovers lots of problems quickly ☺
- But...
  - formal testing sessions are expensive
  - feedback delayed by weeks
  - overkill: far less resources needed to find problems than to fix them
    - What's found in a day is not fixed in a month
- Works with *slow* development methods
  - Huge activity
  - Taken once in a while
  - Produces lengthy documentation
    - largely useless



Poor Man Usability Testing

- … or Agile Usability Testing?
- Simplified version
  - no lab/mirror, just camera
    - both record and streaming to other room
  - three people
    - no elaborate recruiting
  - no report
    - debriefing over lunch
- Extremely low cost
- Can be done frequently
  - e.g. half a day each month



How to Manage Users

read users'minds

observe don't lead

When this happens:	Say this:
You're not absolutely sure you know what the user is thinking (see below).	"What are you thinking?" "What are you looking at?" (for variety) "What are you doing now?" (e.g., if you think they're being silent because they're reading)
Something happens that seems to surprise them. For instance, they click on a link and go "Oh" when the new page appears.	"Is that what you expected to happen?"
They're trying to get you to give them a clue. ("Should I use the ___?")	"What would you do if you were at home?" "What would you do if I wasn't here?"
The participant makes a comment, and you're not sure what triggered it.	"Was there something in particular that made you think that?"
The participant suggests concern that he's not giving you what you need.	"No, this is very helpful." "This is exactly what we need."
The participant asks you to explain how something is supposed to work. ("Do these support requests get answered right away?")	"I can't answer that right now, because we need to know what you would do when you don't have somebody around to answer questions for you. But if you still want to know when we're done, I'll be glad to answer it then."
The participant seems to have wandered away from the task.	"What are you trying to do now?"

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Focus!

- Problem: even the simplified version finds too many problems
- Solution: focus!
  - instruct the observers
  - after each session write down only the three most relevant problems for each user
- Resist temptation to solve easiest first
  - otherwise serious ones will stay there forever
  - exception: so easy that
    - can be fixed by one person in 15'
    - don't need anybody's else approval

Make it a Spectator Sport



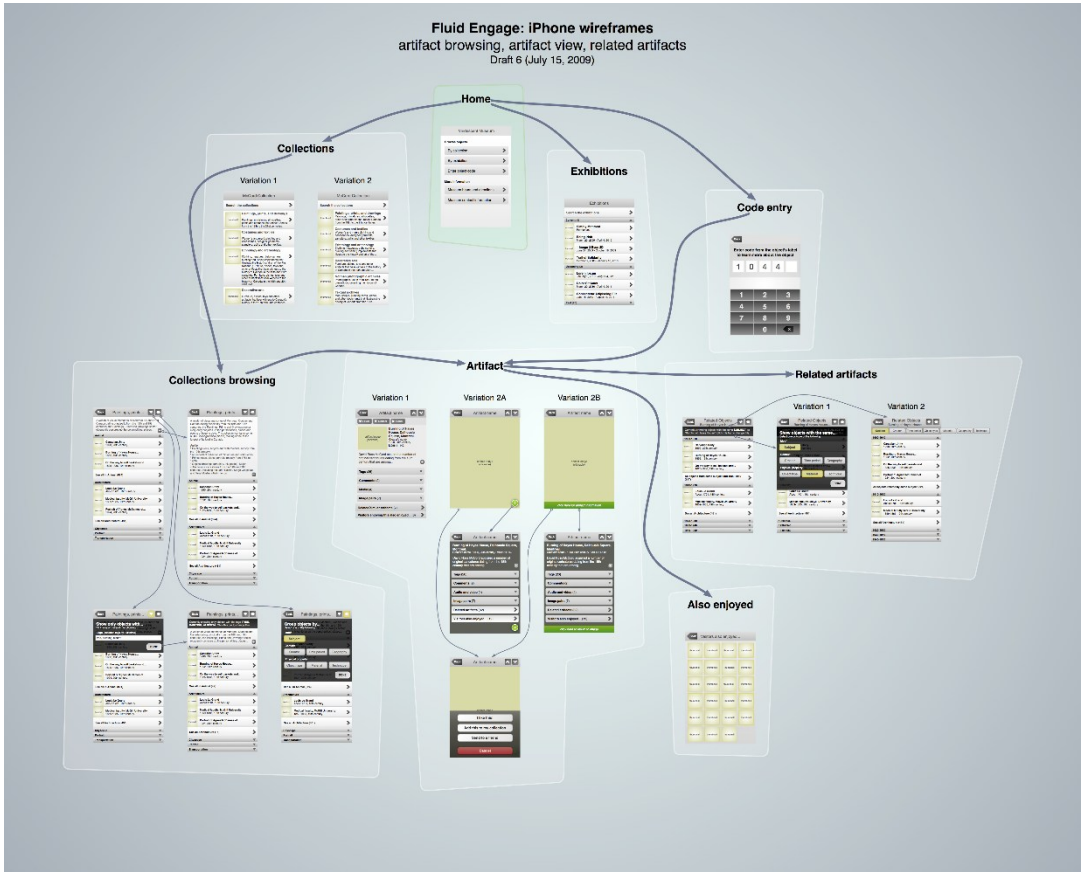
Dibris

Fixing problems

- Always do the least!
- Don’t strive for perfect fix
  - no redesign
  - tweaking
- Tweaking
  - less effort, less time, less cost
  - more likely to happen

When to Test?

- As soon as possible
  - as often as possible
- Test first?
  - preparing task lists
- Paper models
  - less useful if using good developing tool



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